A Report Comparing and Contrasting Techniques for Calibration

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ME-2220 Section Four

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1. Overview

Calibration is an important topic to discuss considering the increasing importance of machines in day to day life. Machines require to be recalibrated at regular intervals, after certain amounts of usage, or when giving non-normal readings. Calibration is key to ensure that machines and devices are operating as designed, thus maintaining reliability and safety. A few common examples are timekeeping pieces, thermometers, and scales. Another important factor to calibrate is the distance between two objects. Two common devices used to measure this distance are a potentiometric displacement transducer (PDT), and the ultrasonic distance sensor (UDS). This report will investigate the calibration process of both, as well as draw conclusions on their real world applications. The PDT uses voltage to measure the distance, and the UDS uses ultrasonic waves to measure distance. The PDT has many uses, but can commonly be found in throttle position sensors in modern vehicles, due to its precision being required to give necessary feedback to the driver through slight changes in the throttle, whereas the UDS is commonly used in robot applications for sensing objects that could be interfering with a path. This report will discuss the two day experimental process in which data was collected through different methods in order to determine the calibration curves for both devices. The report will showcase the results, dive into the uncertainties, and compare and contrast the PDT and UDS.

2. Experimental Methods

There were two different methods of calibration due to the differences in the types of sensors. The first day began with the potentiometric displacement transducer. The materials required for this lab include a multimeter, the PDT, a voltage supply, and a micrometer. The PDT measures distance through voltage by using a strip of electrically resistant material, and a slider. As the slider moves, it changes the location of the ground, and therefore the ground of the circuit.

These changes can be picked up by any tool capable of measuring voltage output. Finally, the micrometer was necessary to accurately calibrate the PDT because it is a precision measurement device with one setback. Due to the gear reduction within the micrometer, measurements must be taken going up in displacement, as well as back down. The slack within the gears of the micrometer, called hysteresis, can lead to disparity between the recordings of distance, and the actual displacement due to the gears not engaging. Both data sets resulting from measuring upscale and downscale are plotted in Figure 1. Twenty measurements were taken with increments of five thousandths of an inch both upscale, as well as downscale, ensuring to rotate past after the final measurement was taken on the upscale to ensure hysteresis does not affect the first downscale data point. With all forty measurements taken, day two begins.

The materials for the second day include an ultrasonic displacement sensor, a breadboard, an LED, a resistor, and an arduino. Using the wiring diagram from the lab manual, the resistor and LED were connected in series on the breadboard. To ensure proficiency with the arduino and software, the first step was to make the LED blink. After the blinking was achieved, the UDS was connected to the arduino through the breadboard via another wiring diagram. The UDS works through a two channel system, one side, the emitter, and the other, the receiver. By knowing when the signal was sent, received, and the speed of sound, the distance of an object can be calculated. The UDS is constantly taking value points so for this experiment, twenty data points were taken representing the most common value shown at every increment of half of an inch. After the distance data, the program was allowed to run and capture 100 data points at a fixed range for use later during the uncertainty calculations. This concludes the experiment portion of the lab.

3. Experimental Results

The raw data from both experiments is included below in Figure 1, and Figure 2. Figure 1 contains both data sets for upscale and downscale. The vertical distance between the two datasets in Figure 1 showcases the hysteresis error between upscale and downscale. The trendline is listed on top of the figures, which also contains the sensitivity as the slope of the line. This will be utilized later for calculating the range.

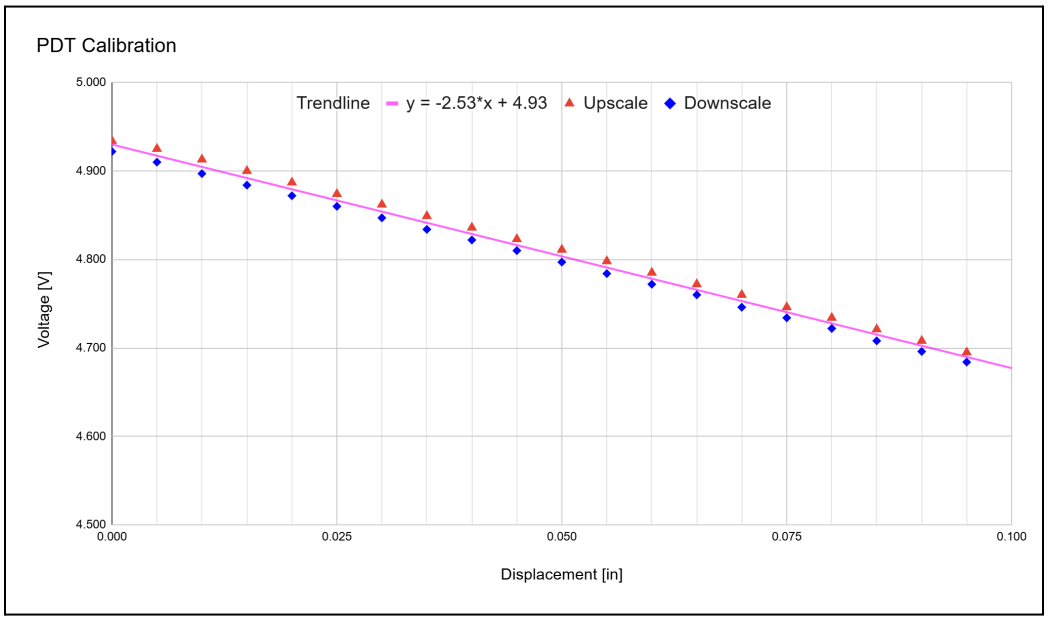


Figure 1: Calibration Graph of PDT

Figure 2 contains a plot with the measured sensing distance on the x-axis, and the time between the sound was emitted, and received. This allows a trendline to be plotted relating the time to the distance allowing the sensitivity to be found through the slope of the line.

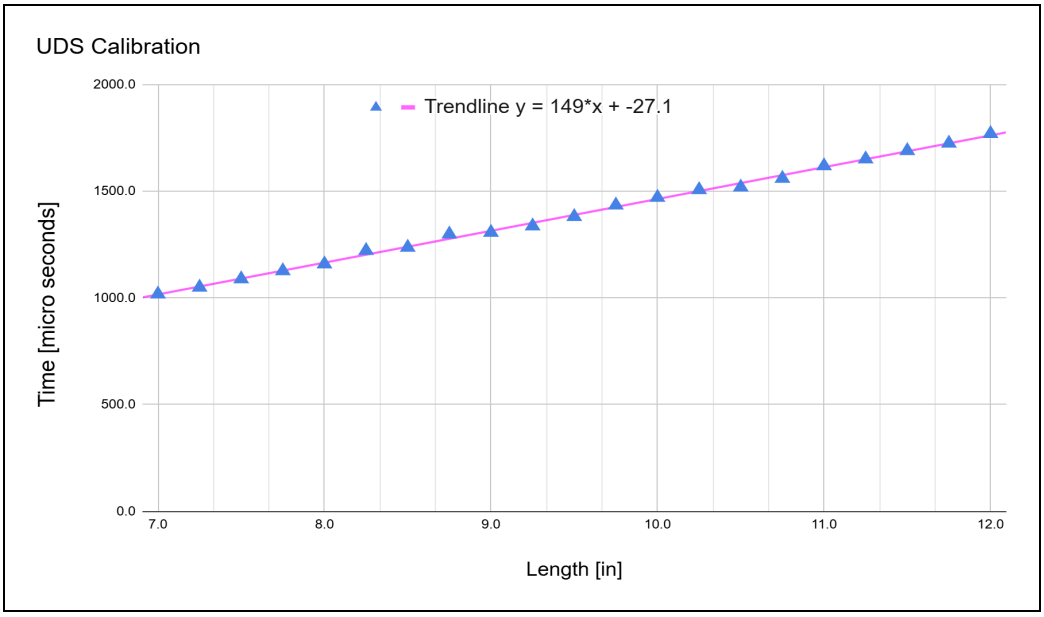


Figure 2: Calibration Plot of UDS

4. Analytical Methods

After the data was plotted, trendlines were created using the equation for linear regression to solve for the slope of the line using all of the data points plotted on the scatter plot. This will be combined with the intercept formula (2) to form the trendline (3).

$$m = \frac{n\Sigma(x_i y_i) - (\Sigma x_i)(\Sigma y_i)}{n\Sigma(x_i^2) - (\Sigma x_i)^2} \quad (1)$$

$$b = \frac{\Sigma y_i - m\Sigma x_i}{n} \quad (2)$$

This equation takes all values and calculates the slope (m) by inputting the dependent variable (y_i), the independent variables (x_i), as well as (n) being the amount of data points used. Now that the slope has been calculated, the intercept (b) can be solved for by plugging in the same variables into equation (3)

$$y = mx + b \quad (3)$$

This equation will be the exact plot of the trendline, and can be seen in both Figure 1, and Figure 2 plotted as the pink line, with the equation stated in the legend. The slope of the linear

regression (m) from equation (1), will be used as sensitivity (S), and combined with V_{max} and V_{min} in order to calculate Range using equation (4). Maximum voltage (V_{max}) and minimum voltage (V_{min}) are the respective maximum and minimum voltage readings of the PDT during the data collection process.

$$\text{Range} = (V_{max} - V_{min})/|S| \tag{4}$$

Linear error (e_{lin}) is an important factor during the uncertainty calculations, and is measured as the difference between the real values, and theoretical values calculated using the trendline. This will be calculated using equation (5).

$$e_{lin} = \left|y_i - y_{ci}\right| \tag{5}$$

Hysteresis error ($e_{hys.}$) is another factor to be considered from the PDT lab, and can be calculated similarly to linear error (5). Instead of real and theoretical values, hysteresis error is calculated by subtracting the upscale values with their corresponding downscale values.

$$e_{hys.} = \left|y_{iup} - y_{idown}\right| \tag{6}$$

Inside of the absolute value, result voltage (y_i) is subtracted by theoretical voltage (y_{ci}) in order to result with the difference between the two. This calculation is done for every change in displacement of both the PDT and UDS. Percent full scale error (%FSE) is computed to show how large the linear error (e_{lin}) (5) is when compared to the range (4) of the data.

$$\%\text{FSE} = \frac{e_{max}}{Range} \times 100\% \tag{7}$$

With the (%FSE) calculated, the total uncertainty (u_{tot}) (14,15) is next. This is calculated using all four sources of uncertainty from various steps throughout the lab. One source of uncertainty is the measuring devices used throughout the lab. ($u_{0,PDT}$) (8) is the uncertainty of the voltmeter, ($u_{0,micrometer}$) (9) is the uncertainty of the micrometer, (u_1) (10) is the first order uncertainty of the

UDS which is solved using (S_x). (S_x) is the standard deviation of the one hundred data points taken from the UDS. Finally ($u_{0,ruler}$) (11) is the uncertainty of the ruler.

$$u_{0,PDT} = \pm\, u_{0,voltmeter} = \pm\, \frac{1}{2}\,[Resolution\ of\ Voltmeter] \tag{8}$$

$$u_{0,micrometer} = \pm\, \frac{1}{2}\,[Resolution\ of\ Micrometer] \tag{9}$$

$$u_1 = \pm\, 2S_x \tag{10}$$

$$u_{0,ruler} = \pm\, \frac{1}{2}\,[Resolution\ of\ Ruler] \tag{11}$$

The remaining uncertainties are hysteresis (u_{hys}) (11), solved using hysteresis error (6), and linear error (u_{lin}) (12), solved using linear error (5).

$$u_{hys.} = \pm\, max(e_{hys.}) \tag{12}$$

$$u_{lin} = \pm\, max(e_{lin}) \tag{13}$$

Finally, total uncertainty can be calculated for both the PDT ($u_{tot,PDT}$) (14), as well as the UDS ($u_{tot,UDS}$) (15).

$$u_{tot,PDT} = \pm\, \sqrt{u_{0,PDT}^2 + u_{0,micrometer}^2 + u_{hys.}^2 + u_{lin}^2} \tag{14}$$

$$u_{tot,UDS} = \pm\, \sqrt{u_1^2 + u_{0,ruler}^2 + u_{lin}^2} \tag{15}$$

5. Analytical Results

The following section will analyze the uncertainties of both the PDT and the UDS in Table 1, and Table 2, before showing the final results in Table 3.

Table 1

Uncertainty Values for PDT

u_{lin} (13) [in]	$u_{hys.}$ (12) [V]	$u_{0,mircometer}$ (9) [in]	$u_{0,PDT}$ (8) [V]	$u_{tot,PDT}$ (14) [%]
0.008	0.016	0.005	0.0164	0.0164

Table 2

Uncertainty Values for UDS

u_{lin} (13) [in]	$u_{1.}$ (10) [ms]	$u_{0,ruler}$ (11) [in]	$u_{tot,UDS}$ (15) [%]
0.145	0.062	0.0625	0.170

Table 3

Resultant Values for PDT and UDS

	Sensitivity (1)	Range (4)	%FSE (7)
PDT	-2.545 V/in	1.9786 in	0.420%
UDS	149.086 ms/in	117.323 in	0.124%

6. Discussion

As can be seen by comparing the total uncertainties from both Table 1 and Table 2, The UDS is an order of magnitude less certain than the PDT. This may also be due to the different factors contributing to the uncertainty formula. The UDS receives comparably more uncertainty from its measurement devices, with the micrometer being a much more certain method of measurement. Even though ignoring the uncertainty contributed through measurement devices, the UDS receives uncertainty from the background noise that was solved with Equation (10). The biggest contributor to uncertainty for the UDT was linear error. This higher uncertainty can

stem from multiple different places, such as the sound waves not bouncing back at the correct angle, or the material absorbing the occasional sound wave. This definitely hurts the UDS sensor, because uncertainty can cause issues if not properly dealt with, but this is where the constant feed of data coming from the UDS is an advantage. Due to the surplus of accurate data the extraneous values are able to be ignored, thus removing this as a limitation. UDS is contrasted by its range being significantly further, extending an extra 115 inches. This gives it many more real world uses and applications. The UDS had a lower %FSE than the PDT, although it should be mentioned that both had very low percentages. The high sensitivity can be seen as an advantage to the UDS, allowing it to record changes in data quickly.

The PDT is an order of magnitude more certain than the UDS. This is most definitely due to the physical connection between the PDT and the micrometer. Unlike the UDS, this ensures that there will not be any extraneous data, and the numbers will be more consistent. While this physical connection allows accuracy, it evidently holds the PDT back when considering common use cases of measuring distance. Hysteresis was one of the contributors to the uncertainty of the PDT, and this is due to the slack within the gear train giving different values depending on the order data was recorded. This can clearly be seen on Figure 1 with the linear error being the measure from the trendline to the plotted points. The sound based UDS does not suffer from this mechanical disadvantage. The relatively higher %FSE seen in Table 3 of the PDT vs the UDS could come from the increased sensitivity of the PDT, making it a more difficult instrument to capture exact data from, due to it being more certain then than the UDS. The PDT also suffers from a lower sensitivity than the UDS, which means that it is not as quick to respond to changes in the displacement, and requires the users to pause and let the multimeter settle on a number

before preceding. By analyzing the data, and comparing the results, conclusions can be drawn about the use cases for both the PDT as well as the UDS.

7. Conclusion

The PDT is a more sensitive device with limited range, which makes it ideal for measuring small distances very precisely, whereas the UDS has a longer range with more uncertainty. The biggest advantage of the UDS is that there is no need to make physical contact with the object you are measuring distance from. This makes it a more ideal device for both longer range, and when physical contact is not feasible. Topics such as these are relevant because there are millions of devices requiring calibration, which are playing important roles such as the previously mentioned throttle position sensor, and automatic doors. Without proper and accurate calibration processes, these could pose a potential safety risk. Both the PDT and UDS are essential components, and each excel in their own respective categories, but neither of these devices are useful without proper calibration from the users. When deciding between these two, a few simple questions can be asked. Is the jump in certainty worth the holdbacks that the PDT brings? Is physical contact possible? If both of these are a yes, then usage of the PDT would be advantageous. If only one or neither are true, then the UDS will suffice for the task.

8. References

Byrd D.S. and Biddlecom J. ME2220 – *Mechanical Engineering Lab I Student Manual Fall 2024* Clemson University.

This lab took a dive into the often hidden, but essential world of PLCs. A simple machine by current standards, the PLC has proved reliable, and robust to last through the years. In addition to the PLC, labor laws were analyzed, and worker productivity was derived from a simple testing procedure. Using these productivity numbers, decisions were made as to what should be prioritized. In the end, a surplus of units was decided, due to the safety cushion that it produces, which allows for the most possibility of error. Trade offs, and decisions like these are made on the daily, and impact countless people. The importance of these decisions can not be overstated, and the people who make them will suffer the benefits as well as the consequences. The PLC was a key to this lab, as well as it will be a key to manufacturing jobs in the field. This lab has demonstrated the criticality of the PLC, and the importance of making a well researched and calculated decision, due to the widespread effects that both have.